Generative Domain Generalization for Autonomous Driving via Diffusion-Based Data Augmentation

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***Abstract*—Domain shift remains a core challenge for deploying autonomous vehicles across diverse environments. Models trained in one geographic region often experience performance degradation in unseen locations due to differences in weather, sensor characteristics, and urban layout. In this work, we present a generative augmentation framework to improve domain generalization in autonomous driving without requiring access to target-domain data. Our method combines diffusion-based stylization and structure-preserving synthesis to simulate realistic distribution shifts from source data. We propose three training-free enhancements—temporal blending, layer-specific gating, and frequency-aware decomposition—to improve visual fidelity and content preservation during synthesis. Experiments on cross-city driving data show that models trained with our synthetic images achieve performance comparable to real target-domain coresets, and slightly outperform source-only and existing generative baselines. Our framework provides a scalable, annotation-free approach to improve the robustness of perception systems in unseen domains.**

I. Introduction

Robust perception in autonomous vehicles demands models that generalize across diverse deployment conditions. However, collecting annotated data across all potential scenarios is impractical. Domain Generalization addresses this challenge by learning from source domains in a way that encourages transferability to unseen ones. While traditional methods rely on architectural regularization or adversarial learning, we instead pursue a generative approach: augmenting training data by simulating domain shifts. Specifically, we leverage DiffuseST [1] for controllable style transfer, guided by BLIP-2 [2] and CLIP [3] embeddings, and integrate ControlNet [4] for fine-grained, structure-preserving generation using geometric priors.

II. Methodology

Our pipeline comprises three stages: stylization via DiffuseST, structure-preserving synthesis with ControlNet, and augmentation-based training. We utilize two pre-trained generative models and condition them on features extracted from established visual backbones.

A. DiffuseST-based Stylization

DiffuseST is a latent diffusion-based style transfer model that injects style semantics into a source image I_c using a reference style image I_s. We extract semantic embeddings via BLIP-2 and CLIP, invert the content image with DDIM, and apply controlled denoising. We introduce three training-free enhancements:

- **Temporal Blending:** A sigmoid-based interpolation controls the gradual style injection across denoising steps. For a normalized timestep x:

$$\begin{cases} y = 1 & \text{if } \sigma(x) \geq 1 - \tau \\ y = 0 & \text{if } \sigma(x) < \tau \\ y = \sigma(x) & \text{otherwise} \end{cases} \tag{1}$$

  where $\sigma(x)$ is the sigmoid function and τ is an empirical threshold (set to 0.01).
- **Layer-Specific Gating:** We modulate the influence of diffusion UNet layers to balance high- and low-level features. Given layer index l:

$$\gamma_l = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \frac{l}{L} \tag{2}$$

  where L is the total number of layers. This preserves layout consistency during style transfer.
- **Frequency-Aware Decomposition:** We blend magnitude spectra of content and style using an adaptive circular mask $M(u, v)$:

$$\begin{aligned} F_b(u,v) &= (1 - M(u,v))\, F_c(u,v) \\ &\quad + M(u,v)\, [w F_s(u,v) + (1-w) F_c(u,v)] \end{aligned} \tag{3}$$

  where F_c and F_s denote content and style spectra, and w is the temporal blending factor.

B. ControlNet Generation

ControlNet enables structure-aware image synthesis by conditioning on priors such as edges, segments, and depth. We utilize: **Canny edges** via the Canny detector [5], **Segmentation masks** from Mask2Former [6], **Depth maps** from MiDaS 3.0 [7]. We explore both single- and multi-modal conditioning for greater control over synthesis fidelity.

III. Experimental Evaluation

We evaluate our method on a cross-city domain generalization task using the nuScenes dataset [8], transferring from Boston (source) to Singapore (target). The evaluation focuses on BEV segmentation performance (mIoU, IoU). We compare five configurations:

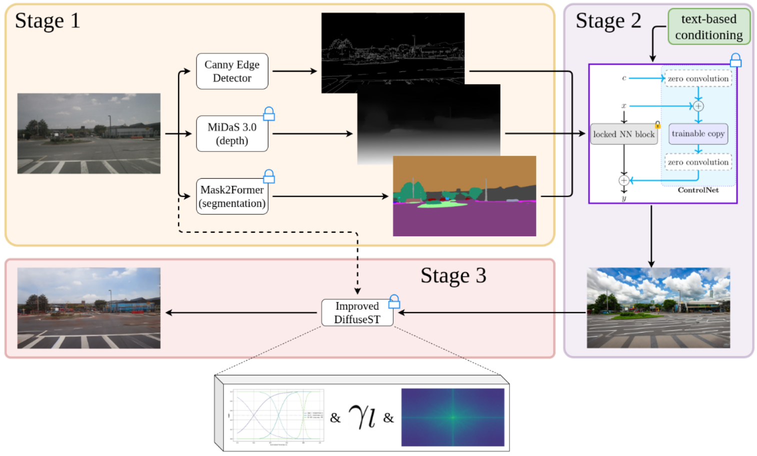

Fig. 1. The proposed three-stage pipeline combining ControlNet and DiffuseST with training-free enhancements.

TABLE I
CROSS-DOMAIN PERFORMANCE ON BEV SEGMENTATION

Configuration	mIoU	IoU
Boston Validation Set	0.3939	0.8279
Singapore Validation (Baseline)	0.2359	0.6893
RC Singapore 200	0.2510	0.6888
AI Studio 200	0.2357	0.6841
ControlNet 200	0.2318	0.6939
ControlNet + DiffuseST Type 1	0.2306	0.6912
ControlNet + DiffuseST Type 2	0.2283	0.6911

1) **RC Singapore 200:** A randomly selected subset of 200 labeled samples from Singapore.
2) **AI Studio 200:** Boston images stylized with Google AI Studio to resemble Singapore.
3) **ControlNet 200:** Structure-aware synthesis from Boston images using semantic/geometric priors.
4) **ControlNet + DiffuseST Type 1:** Full pipeline without frequency-aware decomposition.
5) **ControlNet + DiffuseST Type 2:** Full pipeline with frequency-aware decomposition.

Interestingly, all synthetic augmentation methods maintain comparable IoU to the real-data baselines, with the ControlNet achieving slightly improved IoU over the source-only model on the target domain. While RC Singapore (real labeled target data) offers the highest mIoU, our synthetic variants approach this performance without using target-domain annotations. Notably, ControlNet-generated samples outperform AI Studio's stylizations, and frequency-aware blending (Type 2) maintains competitive IoU while improving visual consistency.

IV. CONCLUSION

We propose a generative framework for domain generalization in autonomous driving that synthesizes diverse scene variants without access to target data. While the performance gains over the baseline are modest, our results show that synthetic augmentation can match real-data coreset baselines in a fully unsupervised setting.

Fig. 2. Augmented images produced by the full pipeline ControlNet + DiffuseST Type 2.

V. ACKNOWLEDGEMENTS

This work was supported by the PNRR MUR project PE0000013-FAIR and developed within the joint project with Leonardo S.p.A.,"Intelligenza Artificiale in sistemi RAdar".

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